

#Multi-LAN Network Architecture with DHCP Configuration (Packet Tracer)#

Introduction:

Objective - This lab demonstrates the design and configuration of a small routed network using Cisco Packet Tracer. The objective was to simulate a realistic enterprise-style network that supports end-device connectivity, inter-network routing, and centralized IP address management using DHCP.

The lab architecture: This consists of three PCs, three laptops, two switches, two routers, and a dedicated server configured to provide DHCP services. End devices are distributed across multiple LAN segments connected via switches, while routers enable communication between the networks.

Focus: The lab focuses on network device interconnection, IP addressing, Subnetting, DHCP configuration, and end-to-end connectivity testing, providing hands-on practice with foundational networking concepts commonly used in real-world environments.

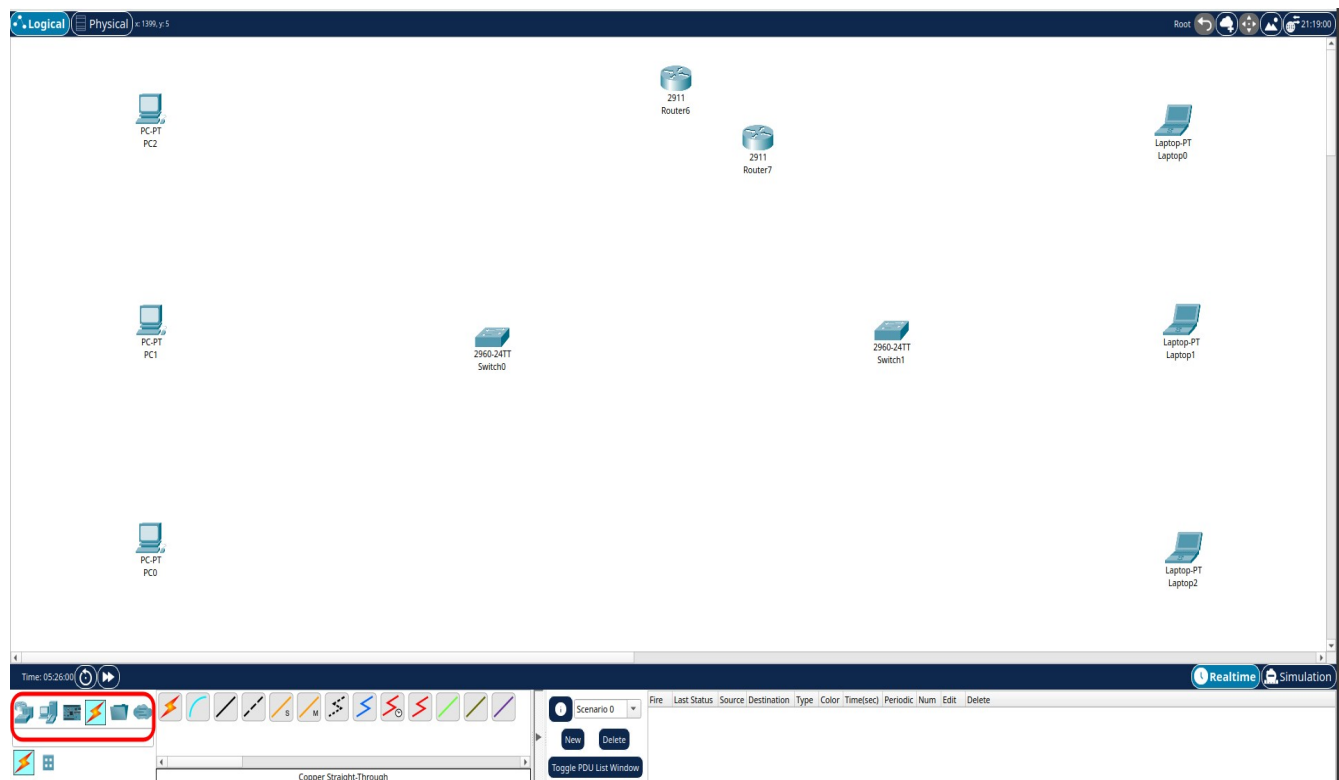
Hands-On

1. Network Topology Description:

Step 1: Devices were sourced from Cisco Packet Tracer.

3 PCs, 3 laptops, 2 switches and 2 Routers

Note: All connections were done using Copper Straight Through Cable



Step 2:

Connecting PC0 – PC2 to Switch0.

PC0:

Click on PC0, select Fa0 and connect it to switch0 by selecting Fa0/1 from switch0

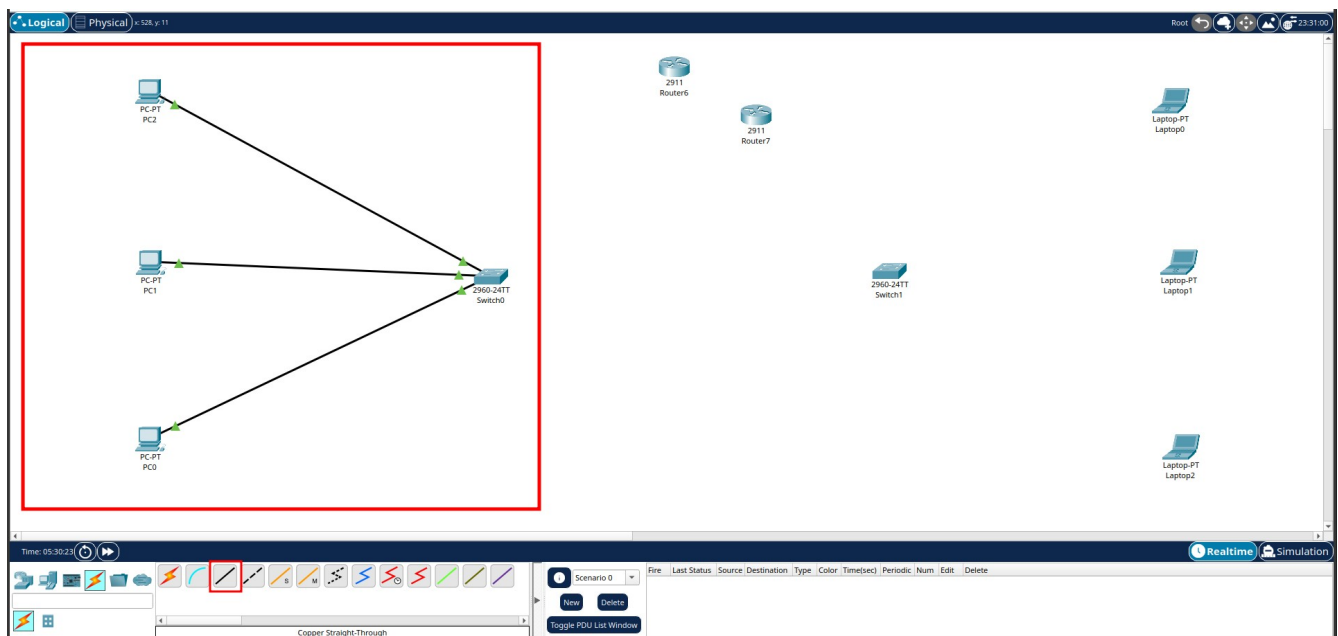
PC1:

Click on PC1, select Fa0 and connect it to switch0 by selecting Fa0/2 from switch0

PC2:

Click on PC2, select Fa0 and connect it to switch0 by selecting Fa0/3 from switch0

Result: Connection successful.



Step 3:

Connecting Laptop0 – Laptop2 to Switch1.

Laptop0:

Click on Laptop0, select Fa0 and connect it to switch1 by selecting Fa0/1 from switch1

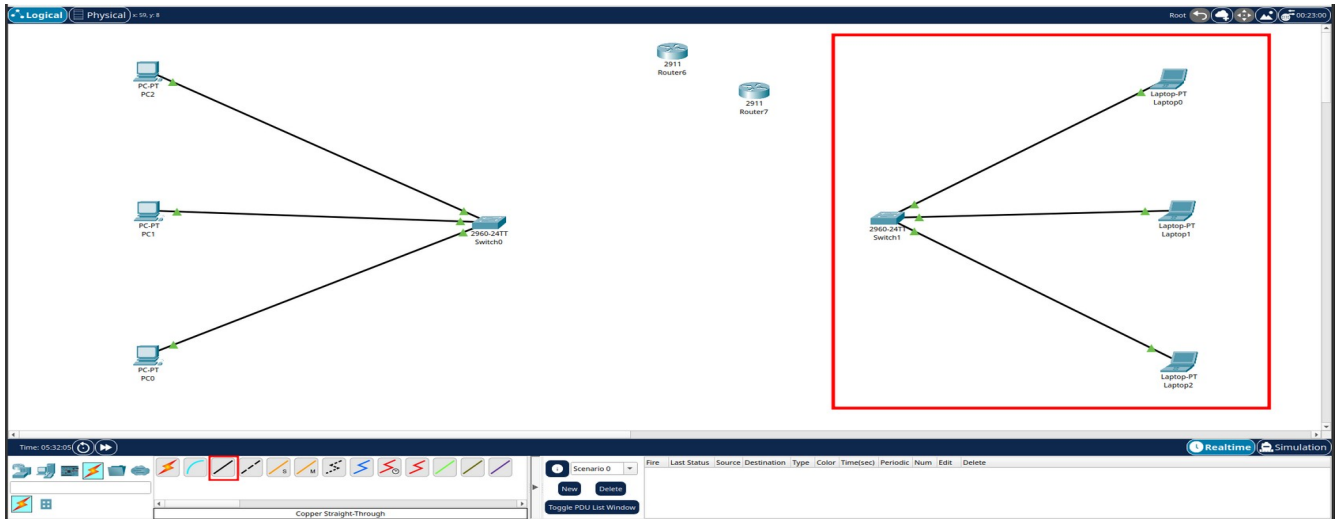
Laptop1:

Click on Laptop1, select Fa0 and connect it to switch1 by selecting Fa0/2 from switch1

Laptop2:

Click on Laptop2, select Fa0 and connect it to switch1 by selecting Fa0/3 from switch1

Result: Connection successful.



Step 4:

Connecting the switches to routers.

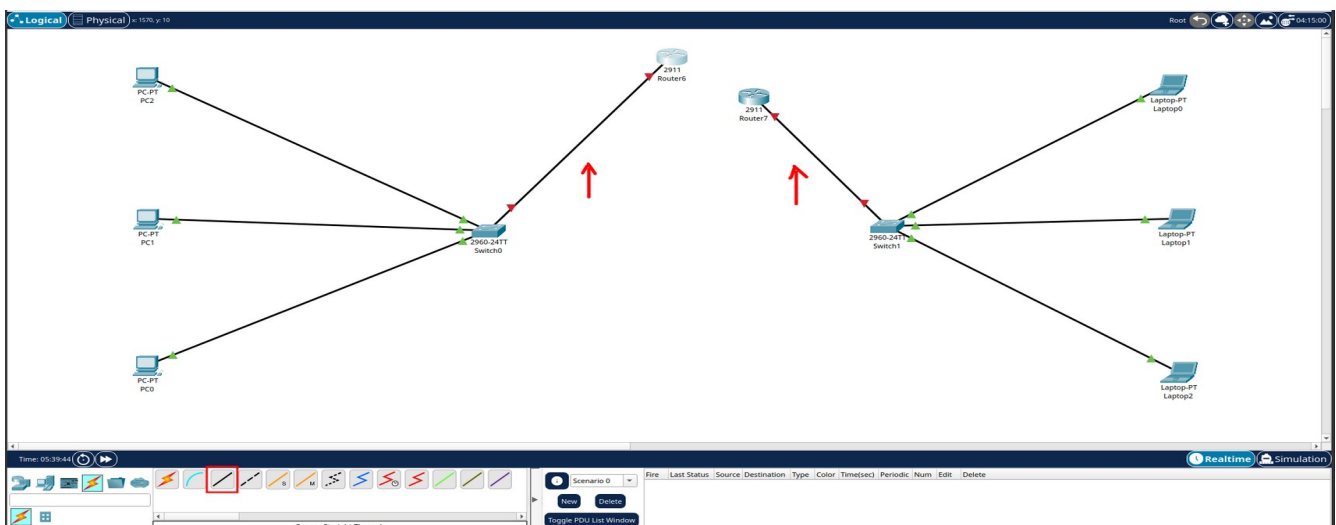
i. Switch0 to Router6:

Click on switch0, select Gi0/1 and connect to Router6 by selecting Gi0/0 from Router6

ii. Switch1 to Router7:

Click on switch1, select Gi0/1 and connect to Router7 by selecting Gi0/0 from Router7

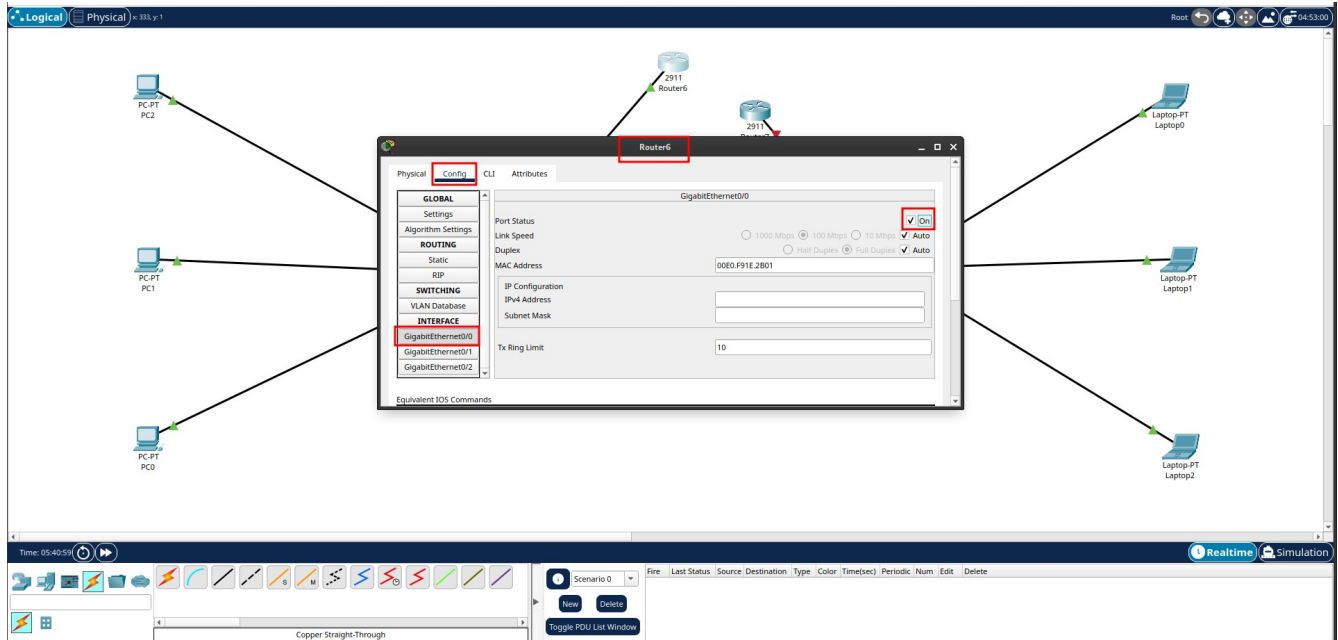
Observation: The connections were not successful because the Ports of Router 6 & 7 are not turned-On hence the red indications on the two connections to the routers6 & 7.



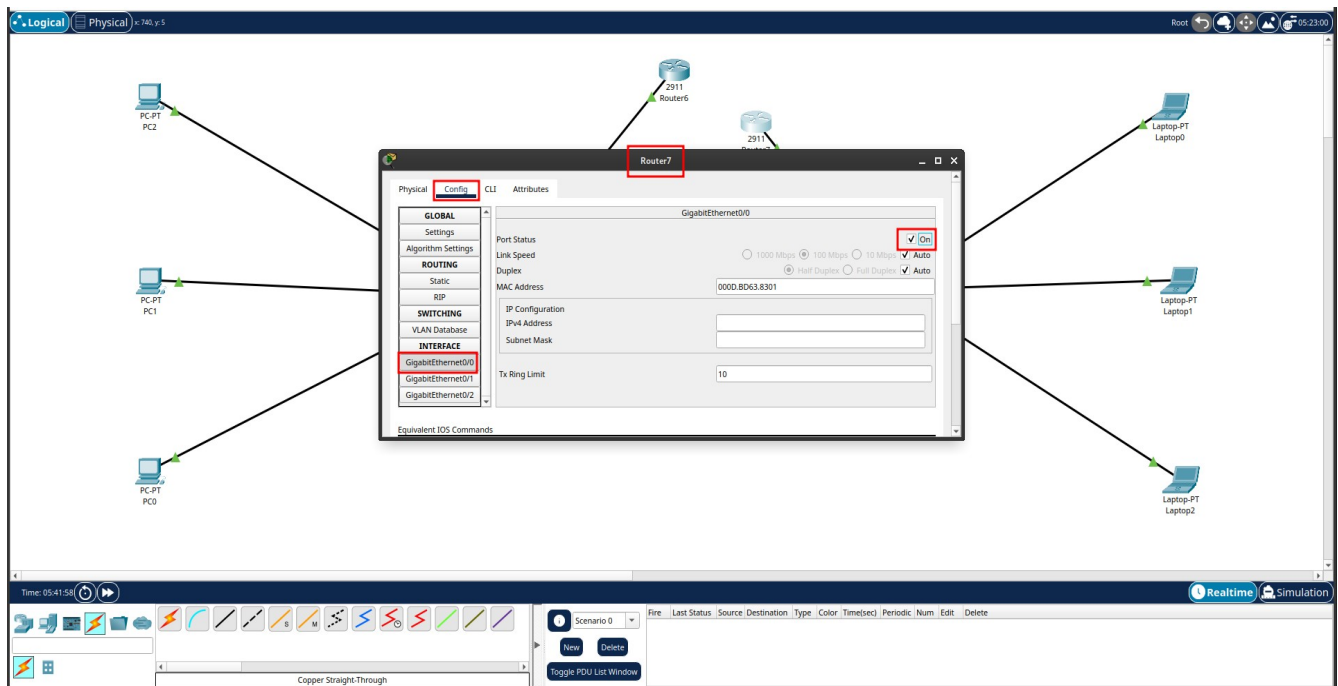
Step 5:

Turning-On the Routers Ports

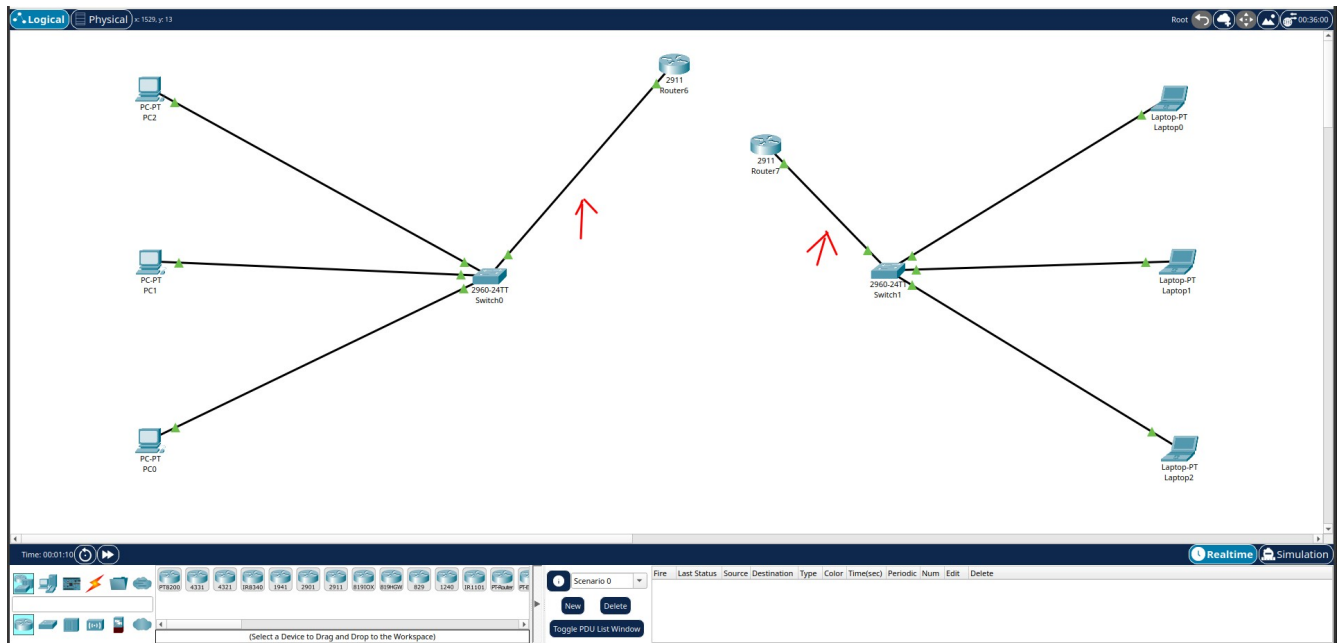
i. Router6 (Conf==>Gi0/0==>On)



ii. Router7 (Conf==>Gi0/0==>On)



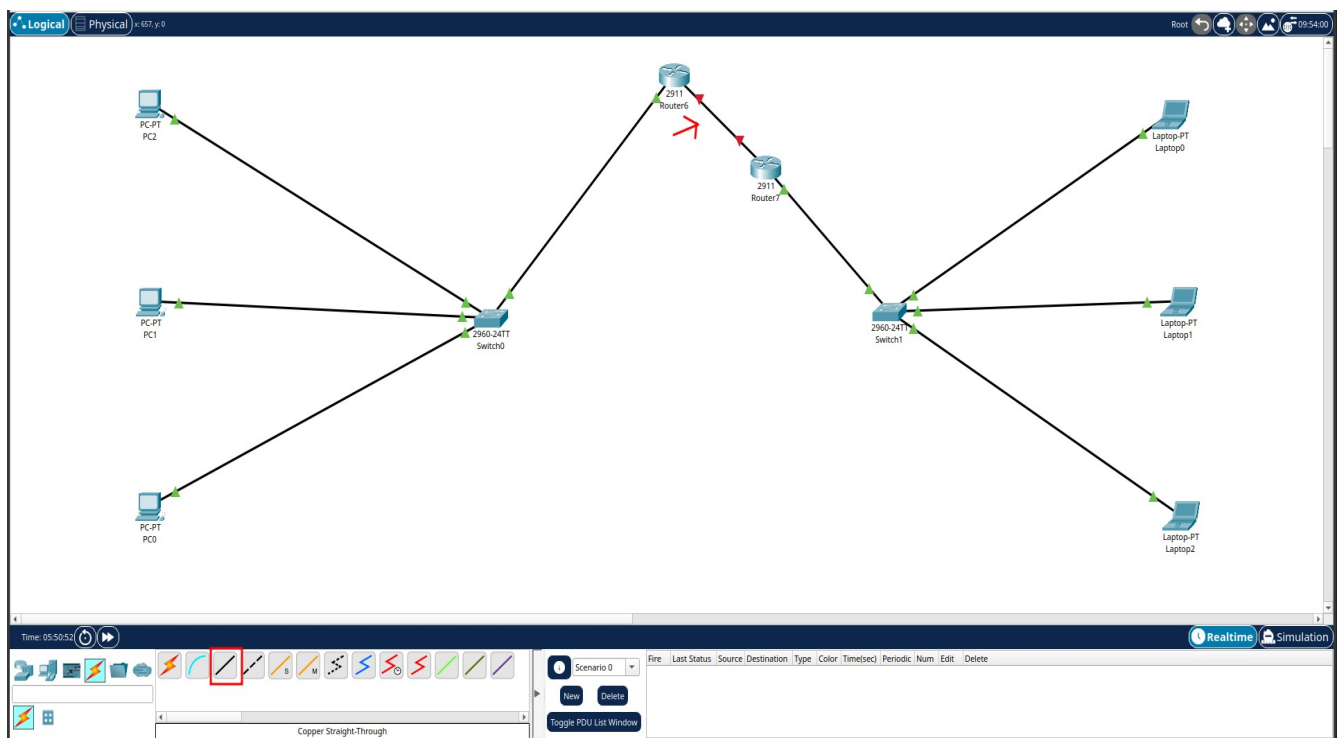
iii. Now Connections are successful as the indicators b/w the 2 switches & routers have turned green



Step 6:

Connecting the two routers (router6 & 7)

Click on router6, select Gi0/1 and connect to router7 by selecting Gi0/1 from router7



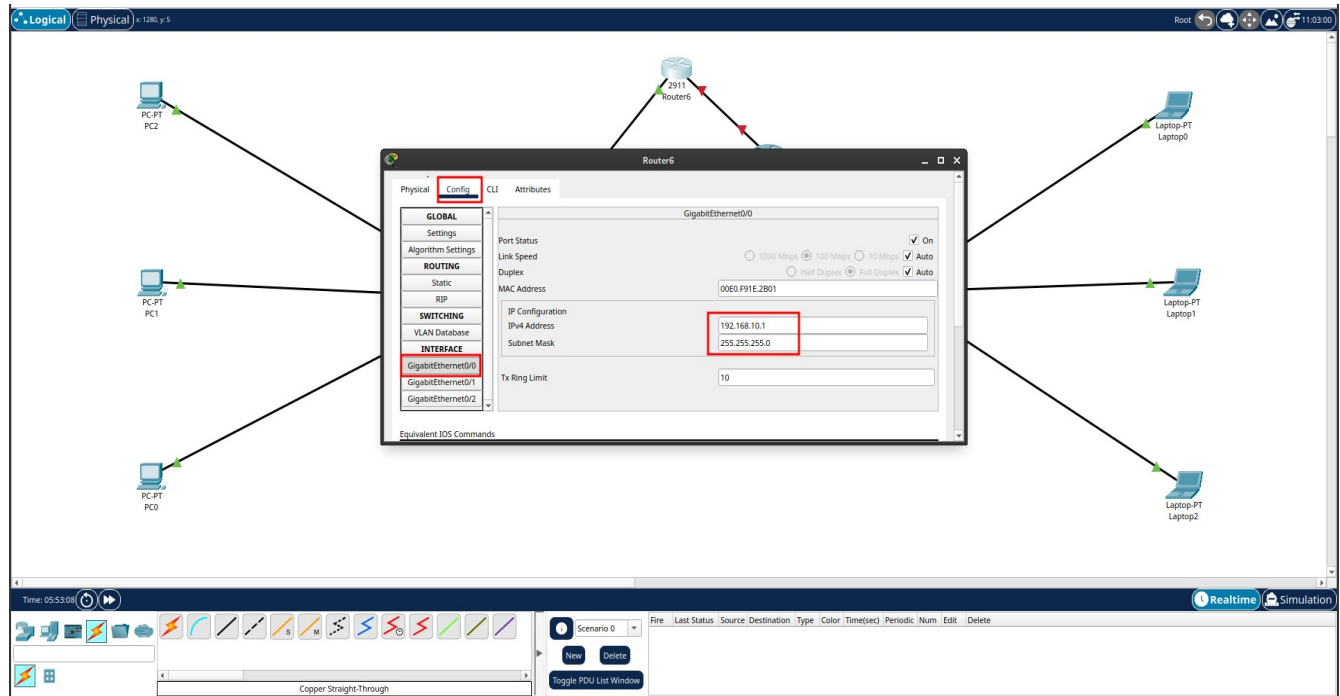
Observation: The connections are not yet successful as there are red indicators shown b/w router6 & 7. This is because, the Gi0/1 ports for routers6 & 7 have not been turned on.

Step 6:

Configuration of Routers (IP addressing & Subnetting)

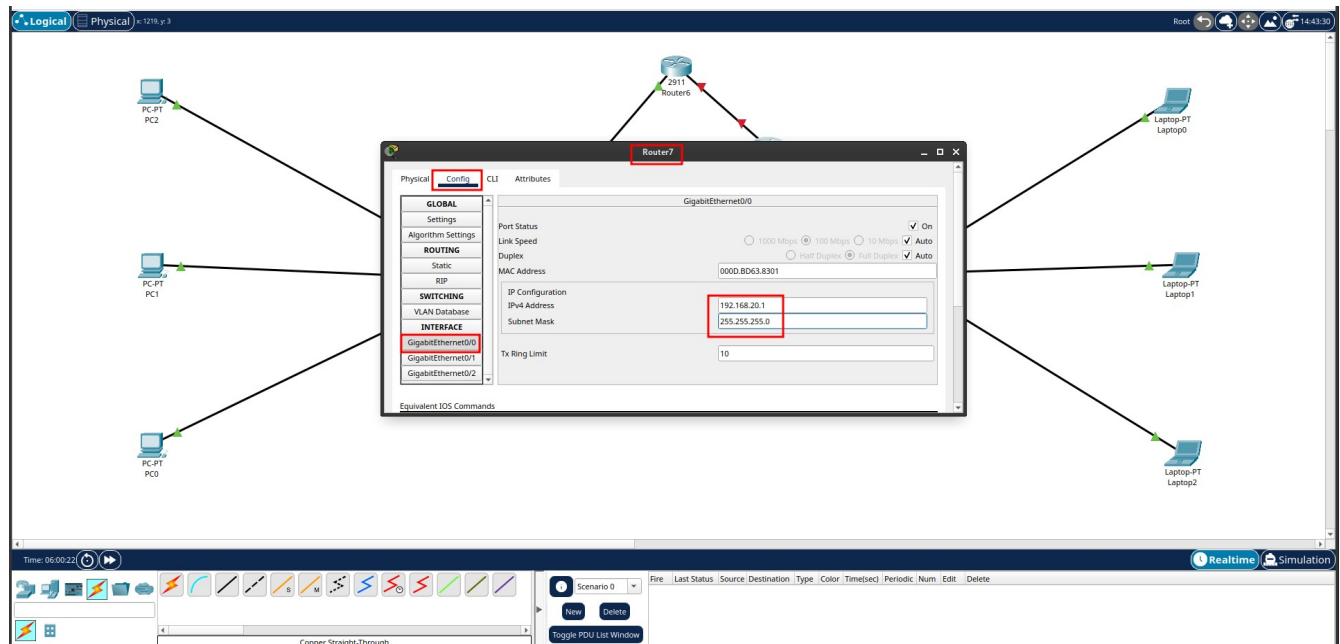
i. Router6 (Gi0/0):

Assigning IP Address (Default Gateway) of 192.168.10.1 with subnet mask of 255.255.255.0 to router6



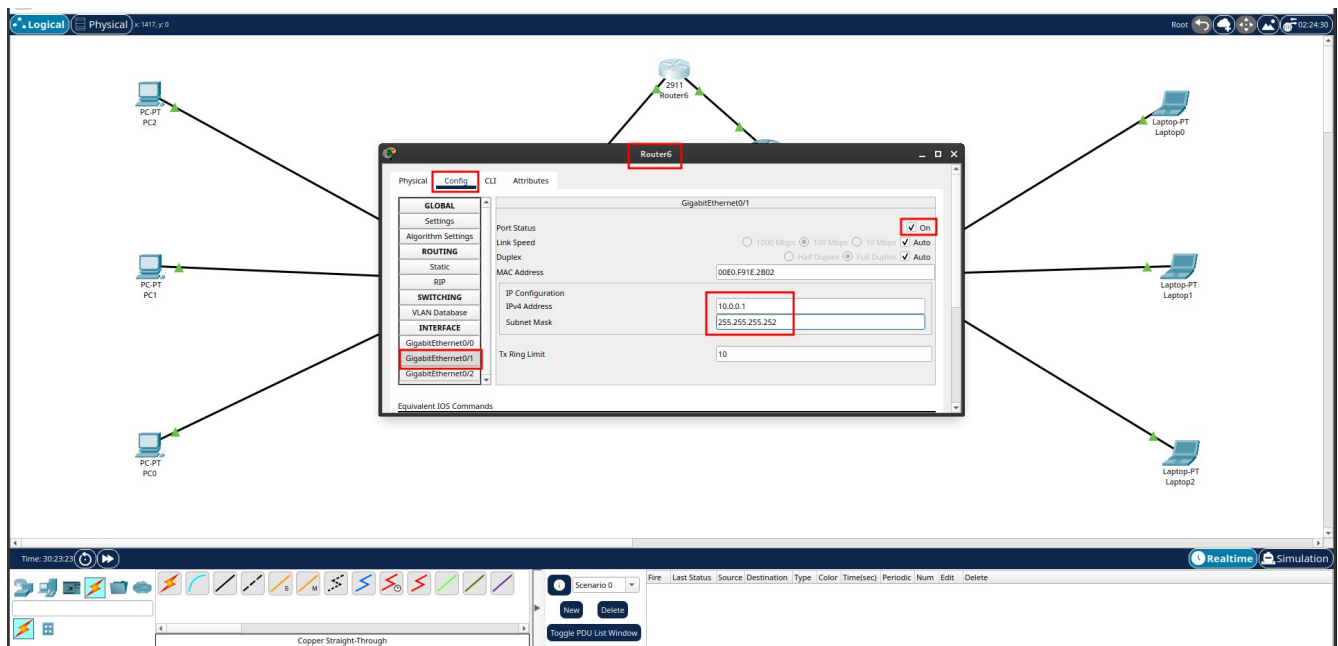
ii. Router7 (Gi0/0)

Assigning IP Address (Default Gateway) of 192.168.20.1 with subnet mask of 255.255.255.0 to router7



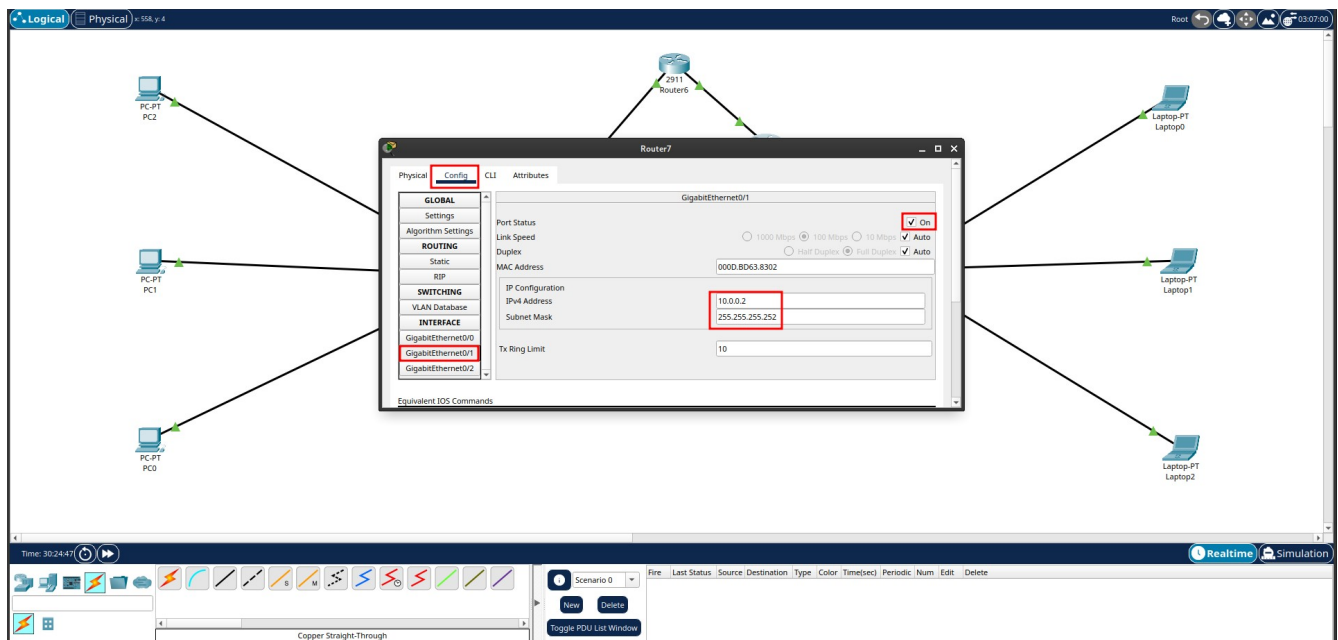
iii. Router6 (Gi0/1):

Assigning IP Address (WAN) of 10.0.0.1 with subnet mask of 255.255.255.252 to router6 and putting the Port ON.

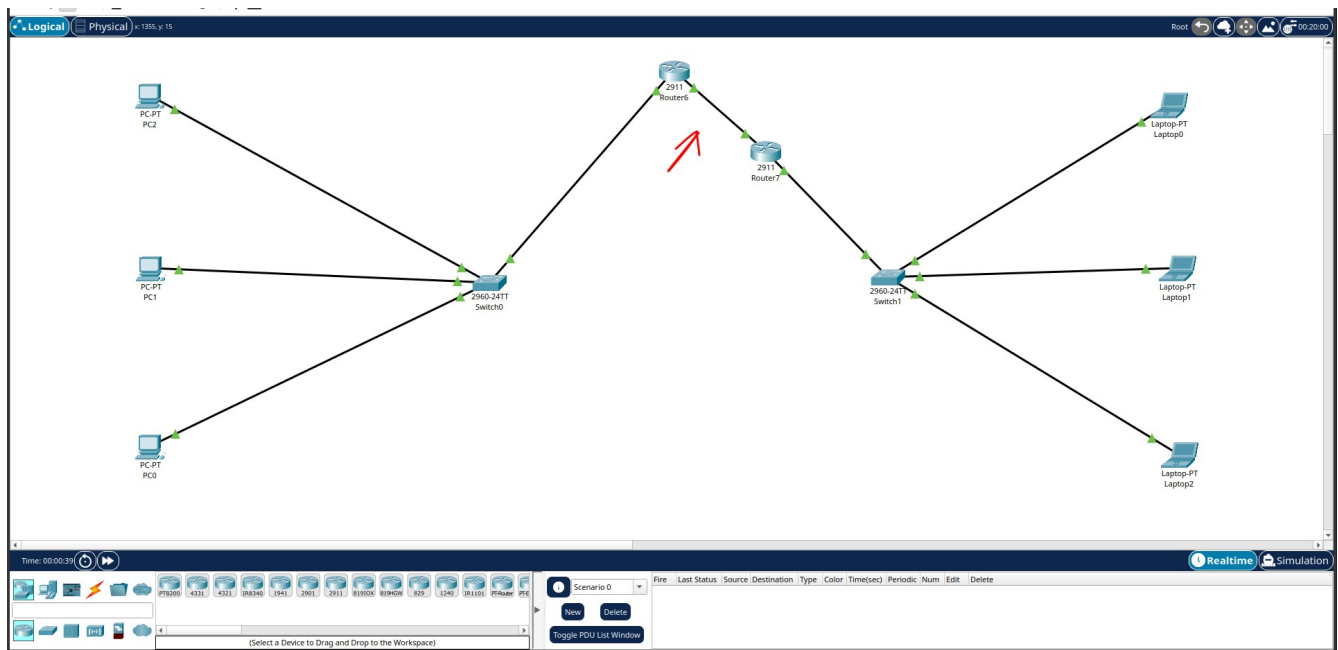


iv. Router7 (Gi0/1):

Assigning IP Address (WAN) of 10.0.0.2 with subnet mask of 255.255.255.252 to router7 and putting the Port ON.



Observation: The connection between the two routers are now successful as the connection indicators b/w the two have turned green.

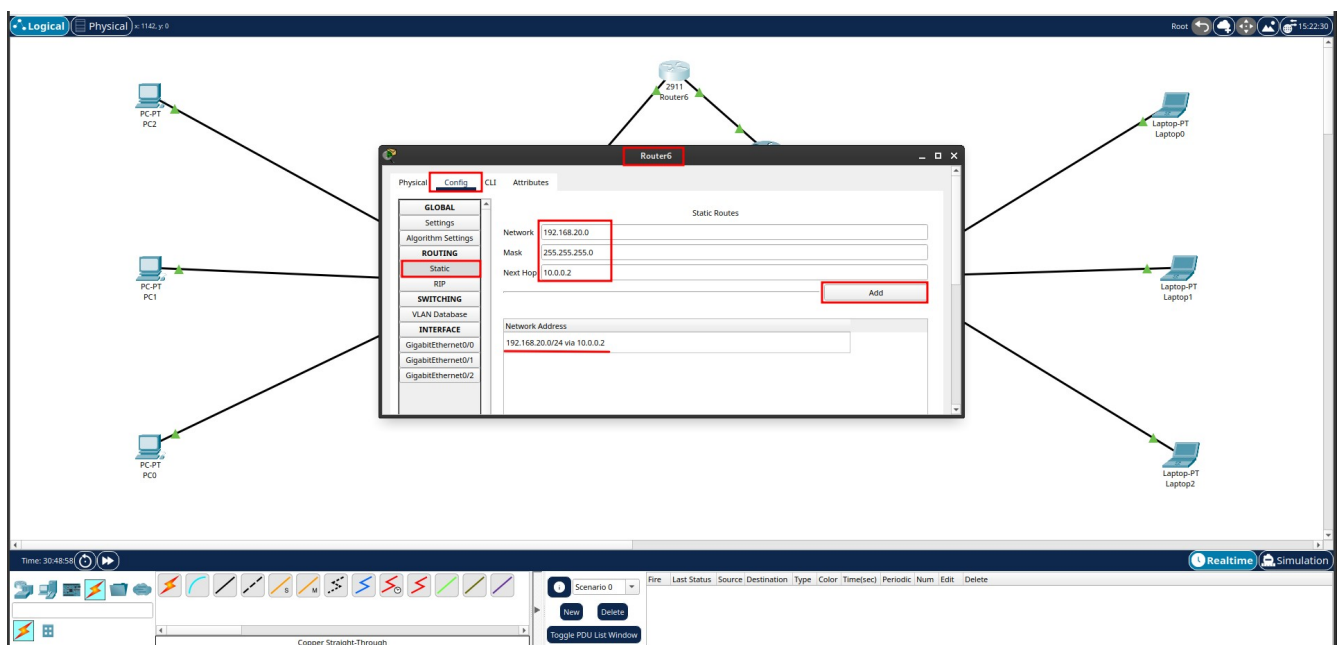


Step 7:

Routing Configuration (Bidirectional Static Routing)

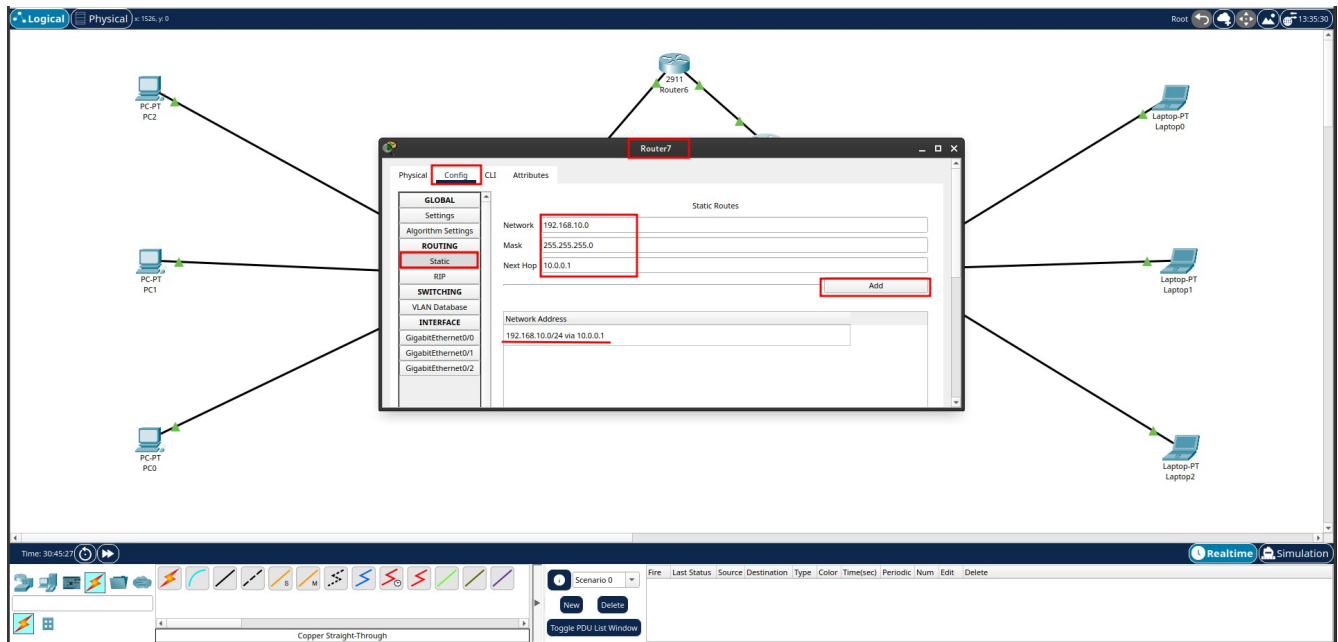
i. Router6

Entering network information of Router 7 into Router 6 and clicking “Add”



ii. Router7

Entering network information of Router 6 into Router 7 and clicking “Add”

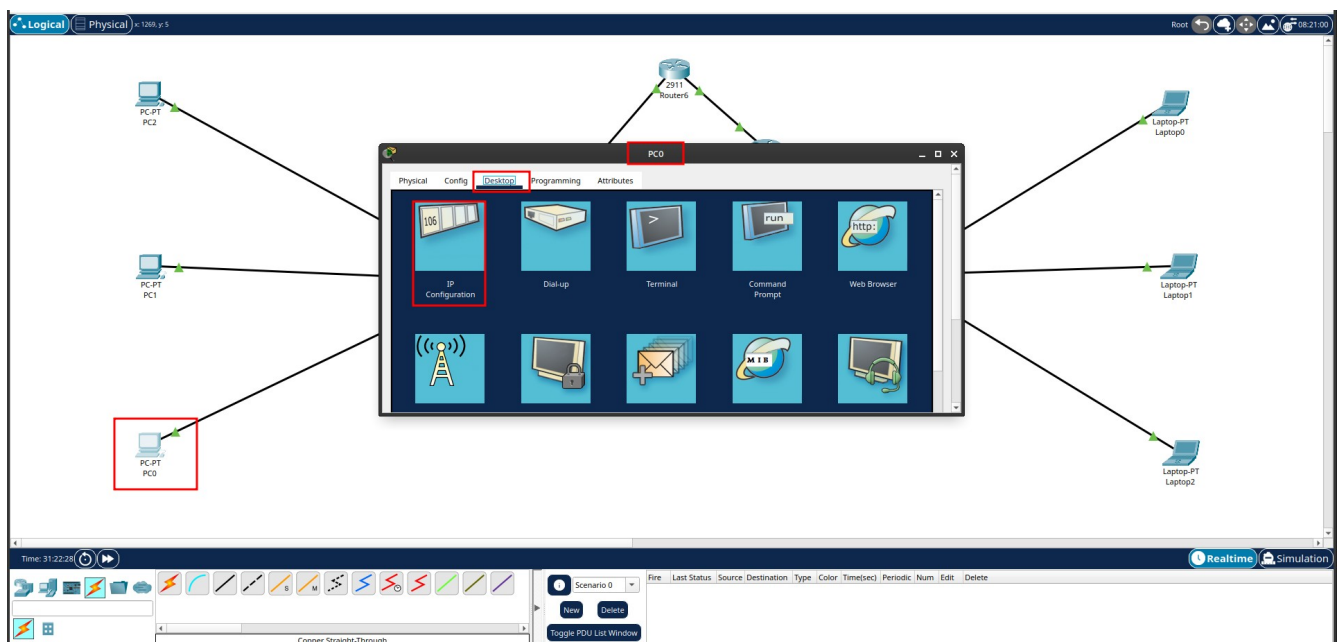


Step 8:

Endpoint Configuration (PC0 – 2 & Laptop0 – 2)

i. PC0:

Click on PC0, click on Desktop and then click on IP configuration

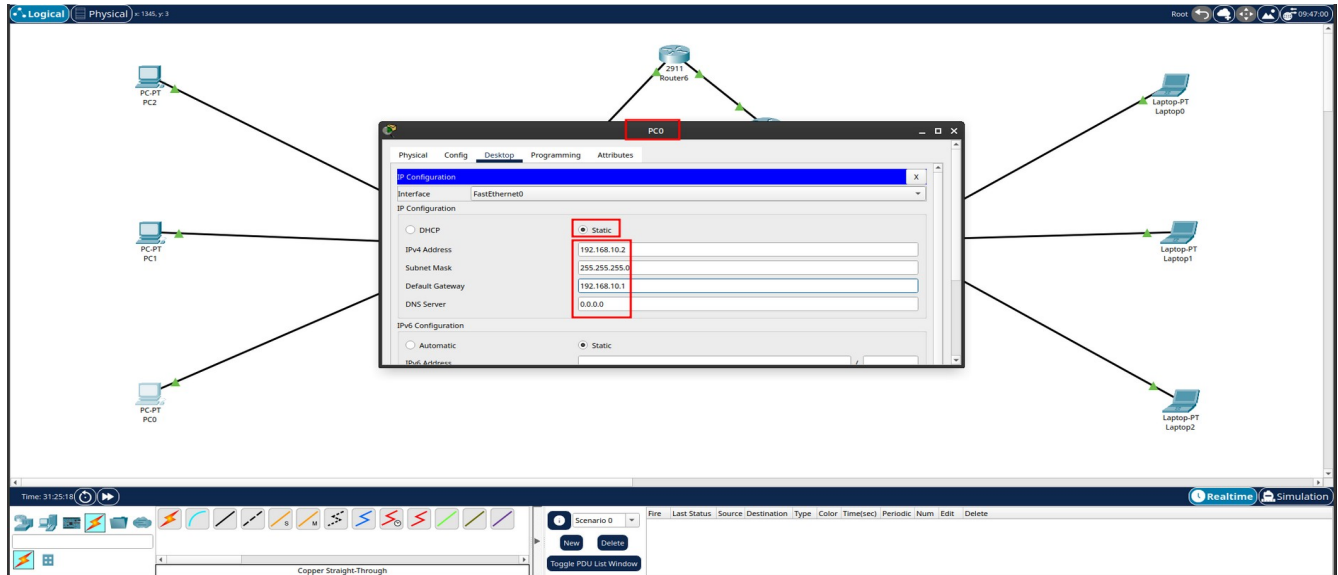


Assigning IP to this device (Host)

IP: 192.168.10.2

Subnet mask: 255.255.255.0

Gateway: 192.168.10.1



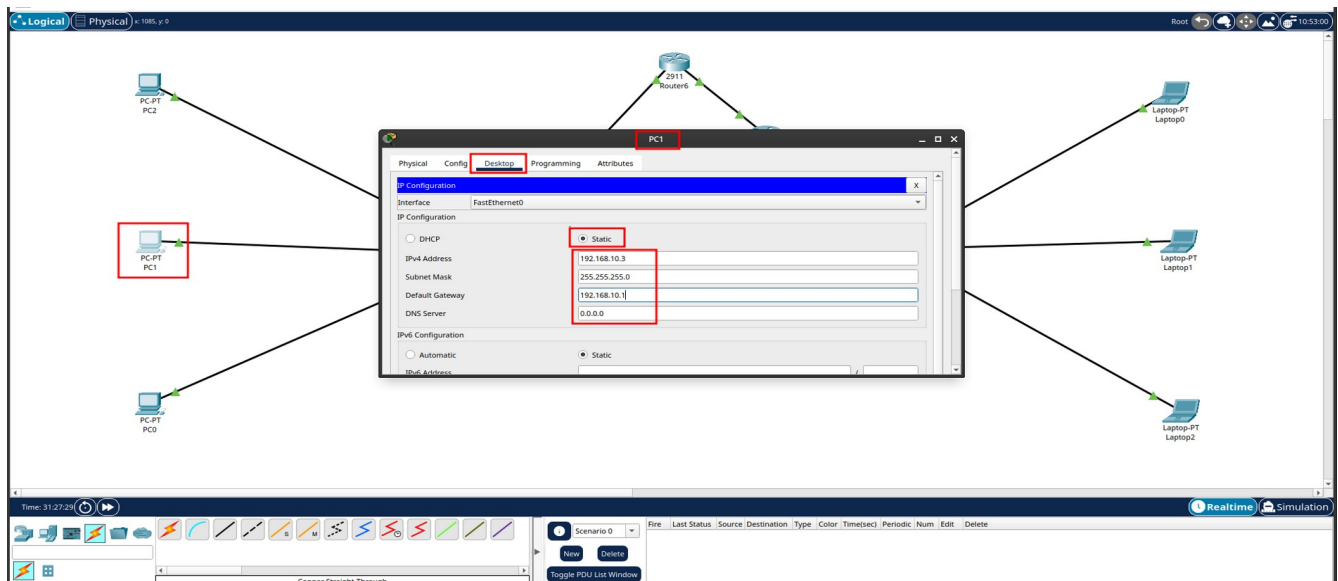
ii. PC1:

Assigning IP to this device (Host)

IP: 192.168.10.3

Subnet mask: 255.255.255.0

Gateway: 192.168.10.1



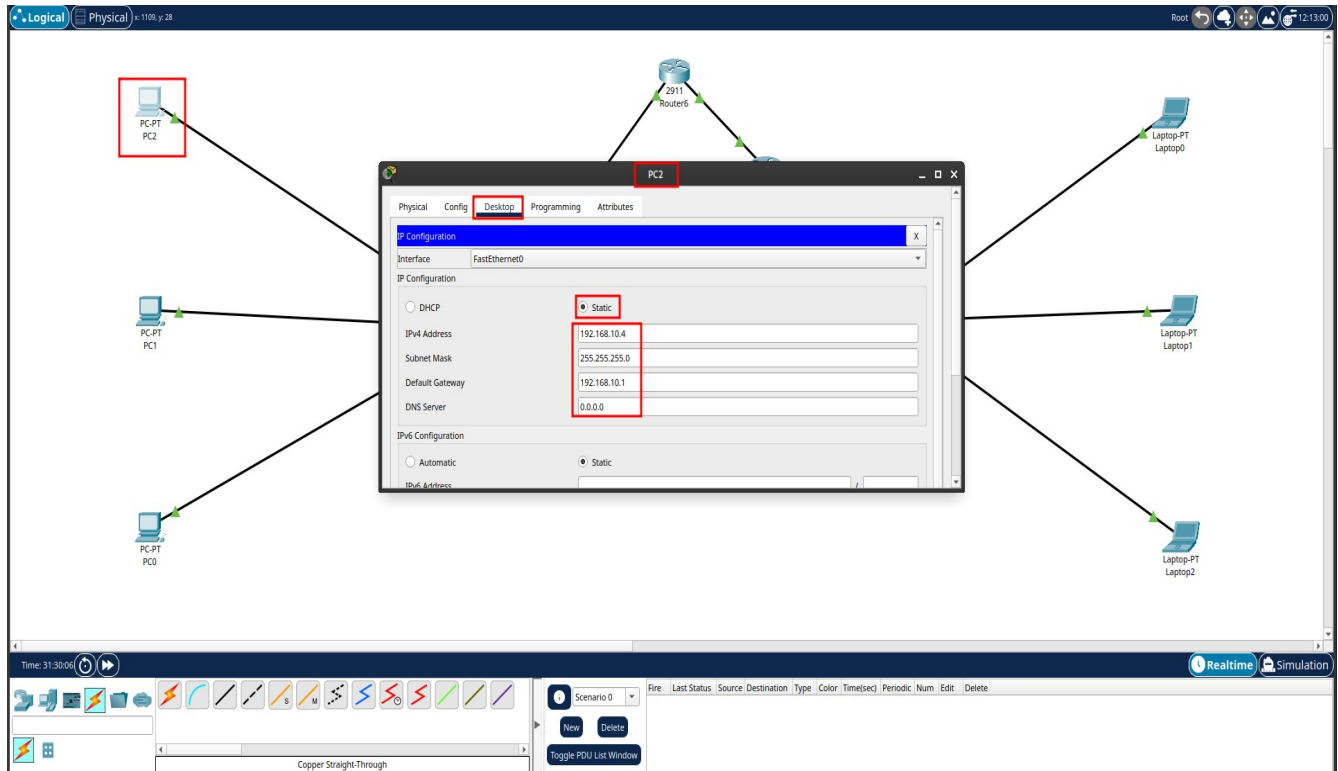
iii. PC2:

Assigning IP to this device (Host)

IP: 192.168.10.4

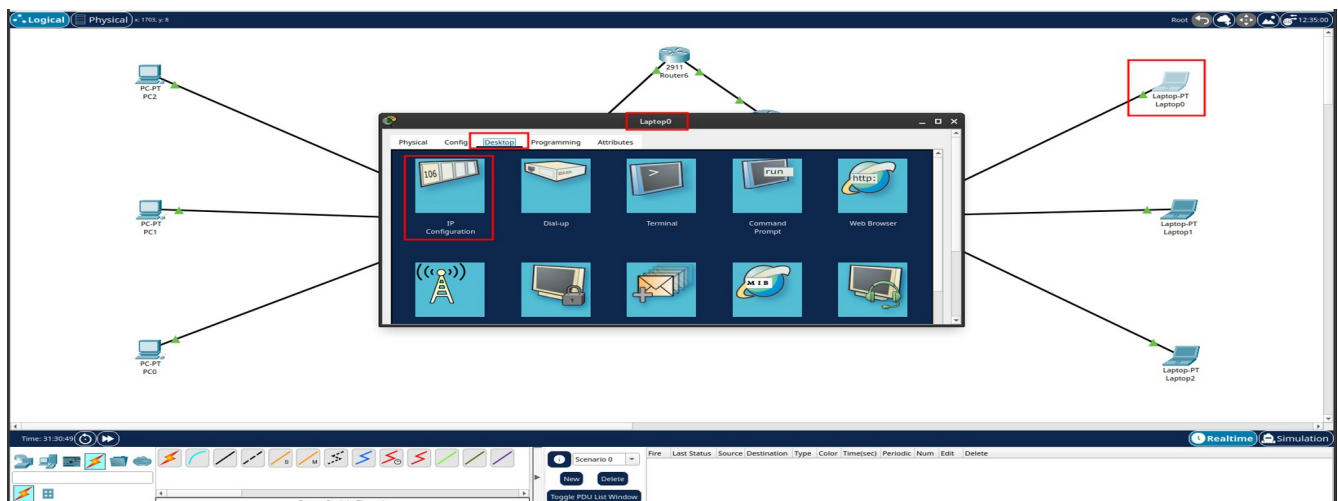
Subnet mask: 255.255.255.0

Gateway: 192.168.10.1



iv. Laptop0:

Click on Laptop0, click on Desktop and then click on IP configuration

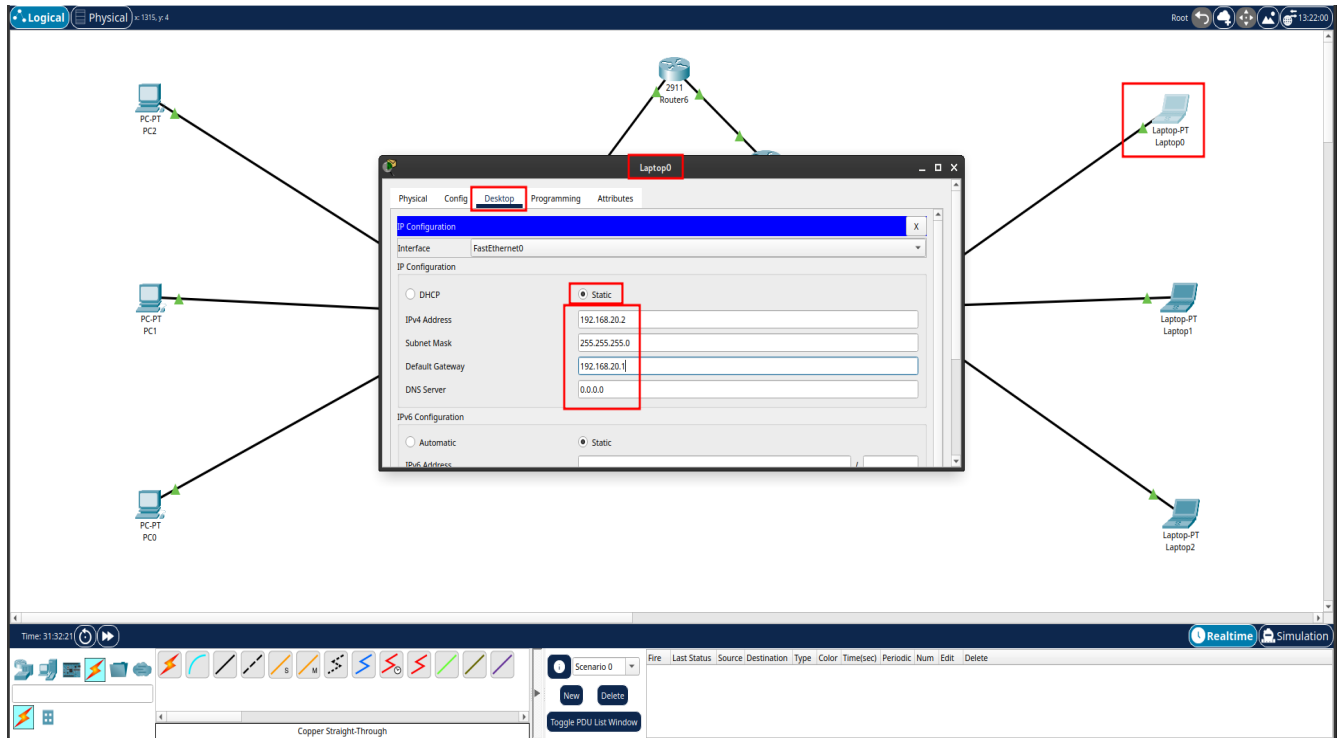


Assigning IP to this device (Host)

IP: 192.168.20.2

Subnet mask: 255.255.255.0

Gateway: 192.168.20.1



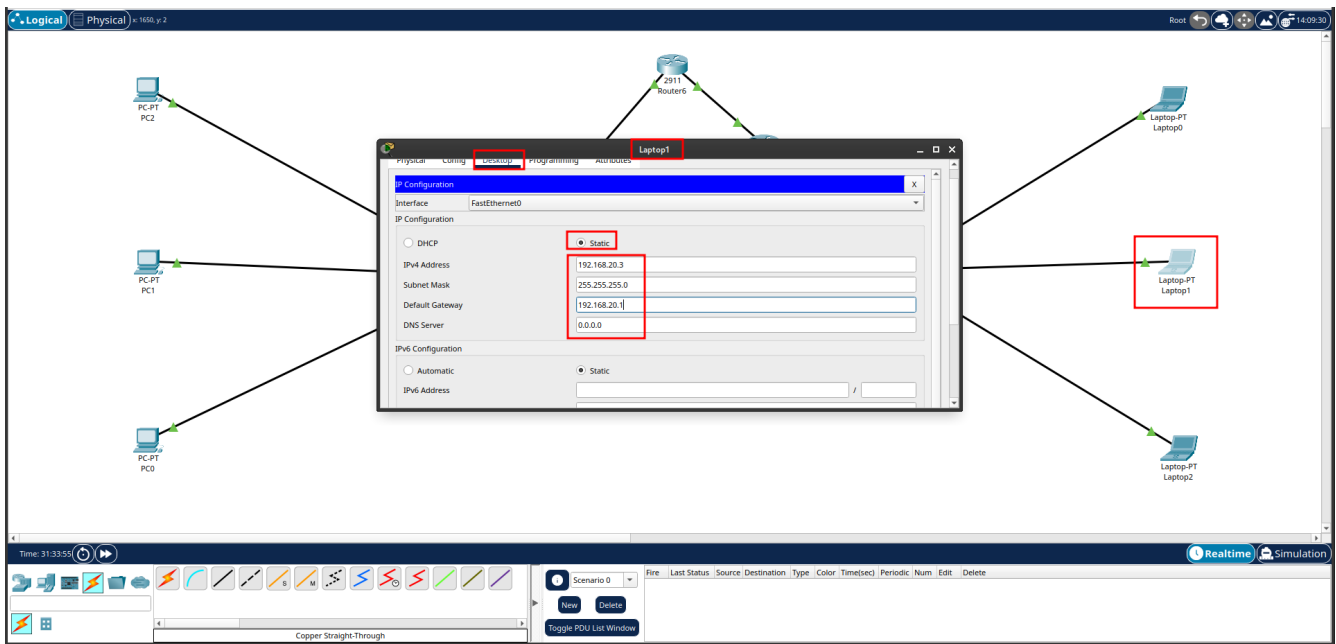
v. Laptop1

Assigning IP to this device (Host)

IP: 192.168.20.3

Subnet mask: 255.255.255.0

Gateway: 192.168.20.1



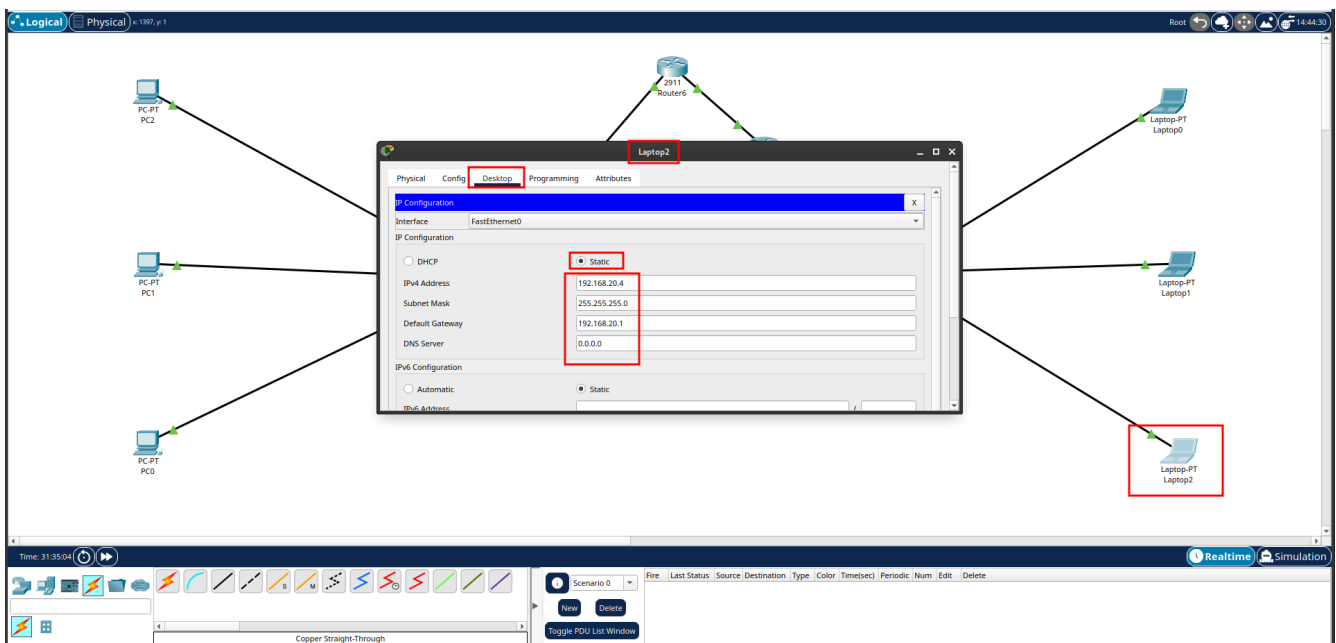
vi. Laptop2

Assigning IP to this device (Host)

IP: 192.168.20.4

Subnet mask: 255.255.255.0

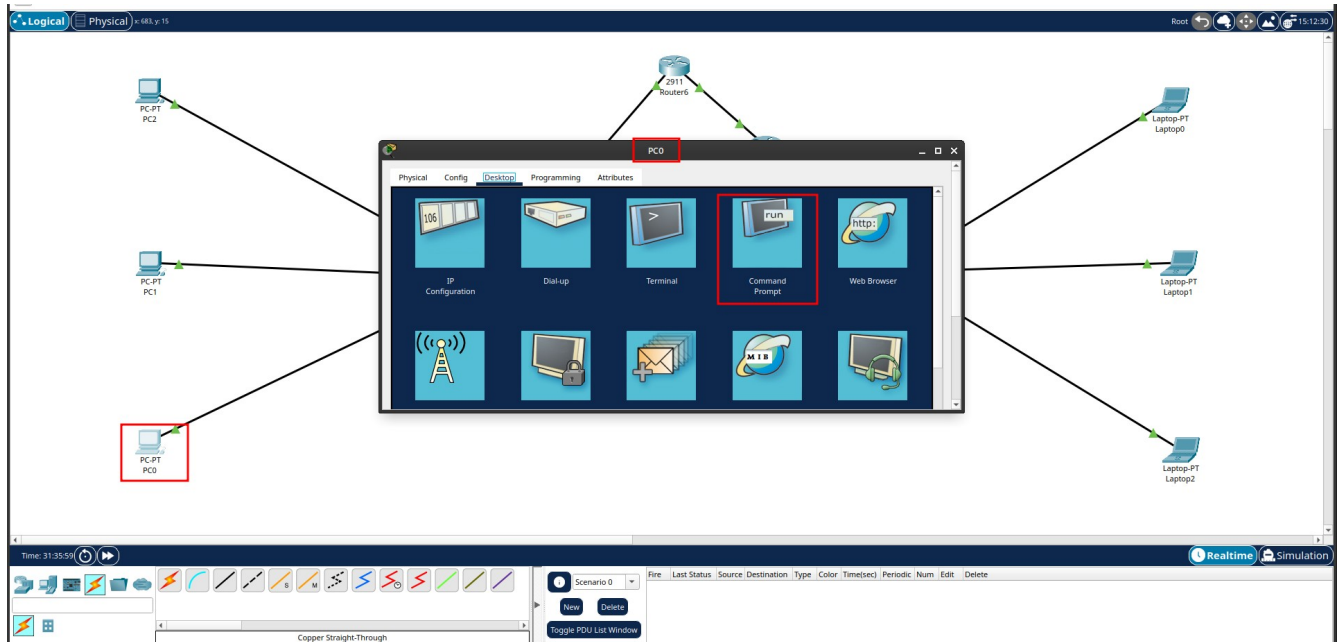
Gateway: 192.168.20.1



Step 9:

Verification & Connectivity Testing:

i. From PC0, ping the local Gateway - 192.168.10.1



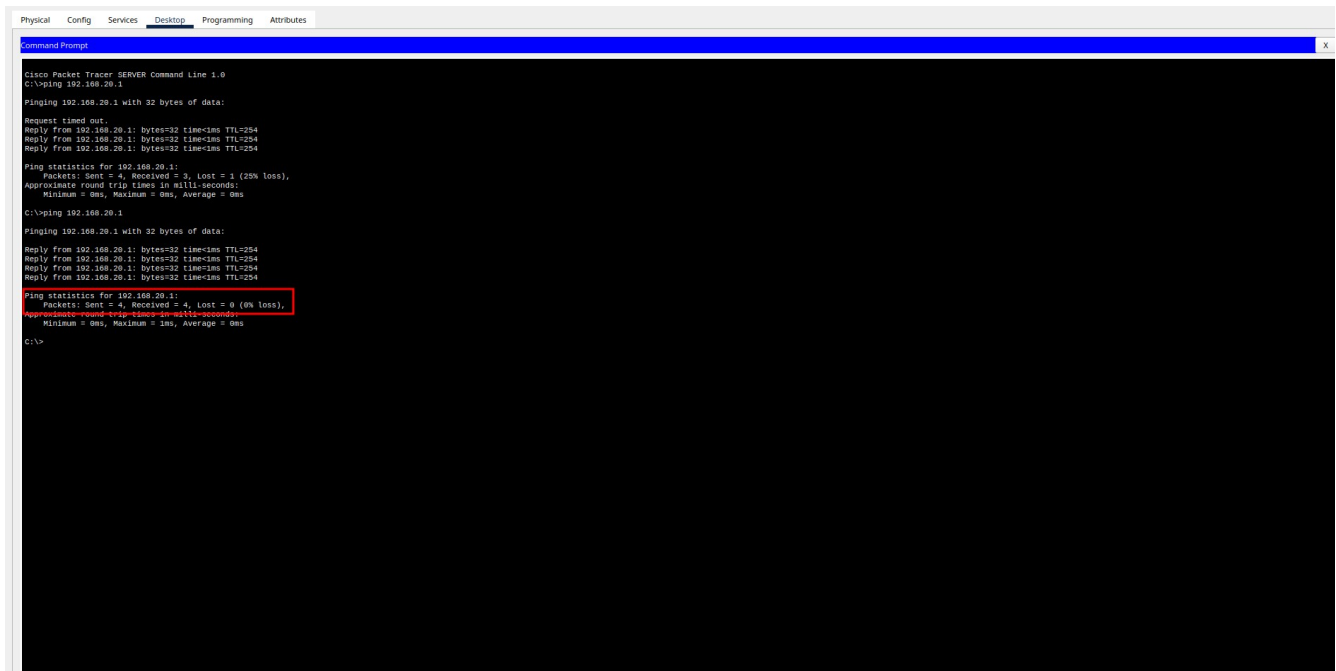
ii. Result:

Packets sent =4, Received =4, Lost =0 (0% loss)

```
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
```

ii. Also, from the same PC0, ping the Neighbour's router7 - 192.168.20.1

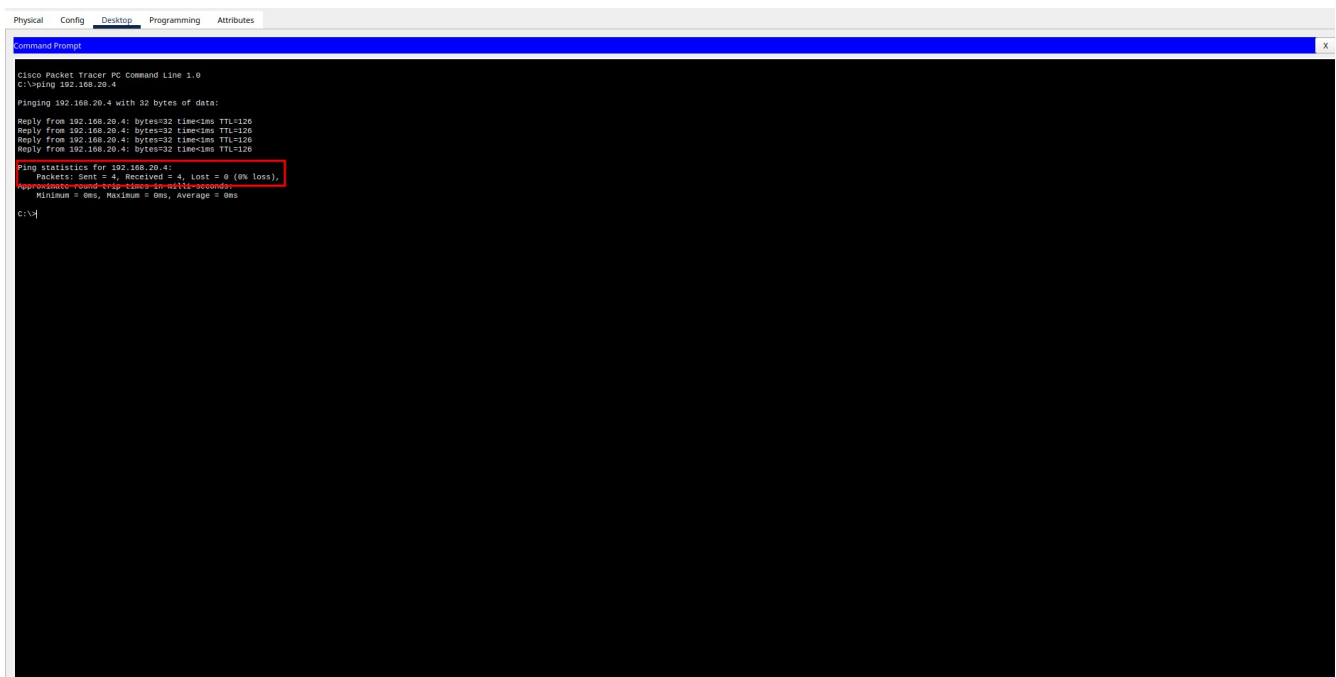
Result: Packets sent =4, Received =4, Lost =0 (0% loss)



```
Physical  Config  Services  Desktop  Programming  Attributes
Command Prompt
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.20.1
Pinging 192.168.20.1 with 32 bytes of data:
Request timed out.
Reply from 192.168.20.1: bytes=32 time=1ms TTL=254
Reply from 192.168.20.1: bytes=32 time=1ms TTL=254
Reply from 192.168.20.1: bytes=32 time=1ms TTL=254
Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.20.1
Pinging 192.168.20.1 with 32 bytes of data:
Reply from 192.168.20.1: bytes=32 time=1ms TTL=254
Reply from 192.168.20.1: bytes=32 time=1ms TTL=254
Reply from 192.168.20.1: bytes=32 time=1ms TTL=254
Reply from 192.168.20.1: bytes=32 time=1ms TTL=254
Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
```

iii. Also, from the same PC0, ping the Neighbour's host - 192.168.20.4

Result: Packets sent =4, Received =4, Lost =0 (0% loss)

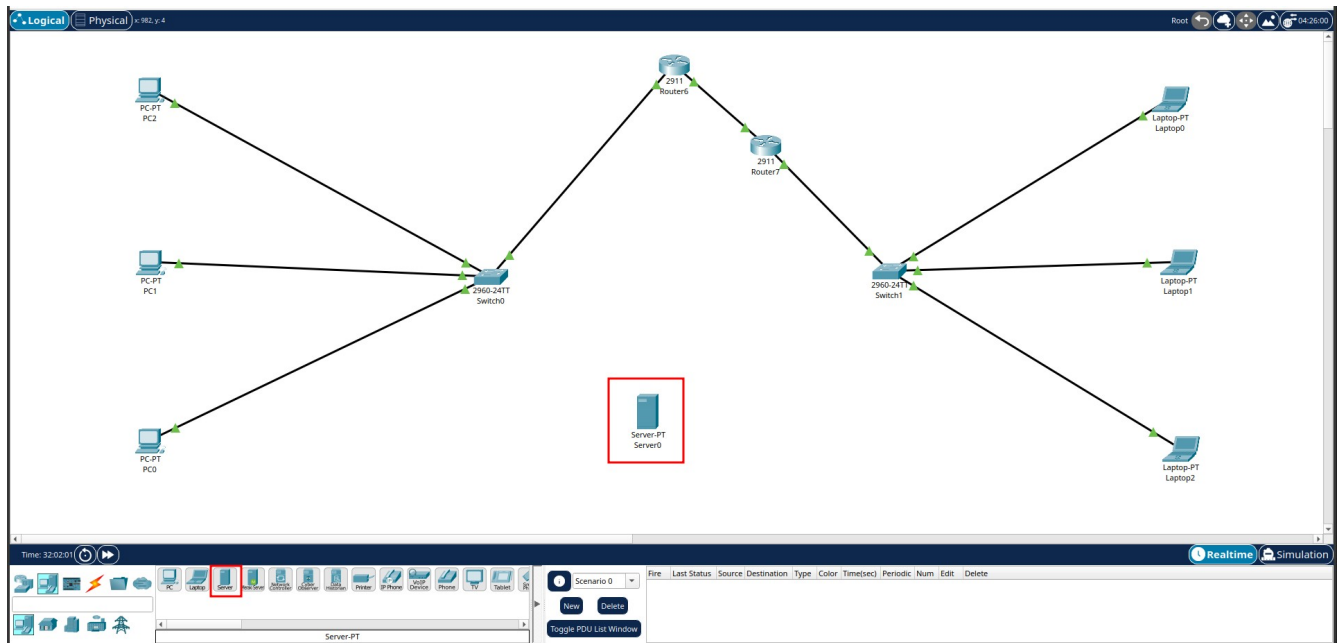


```
Physical  Config  Desktop  Programming  Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.4
Pinging 192.168.20.4 with 32 bytes of data:
Reply from 192.168.20.4: bytes=32 time=1ms TTL=120
Reply from 192.168.20.4: bytes=32 time=1ms TTL=120
Reply from 192.168.20.4: bytes=32 time=1ms TTL=120
Reply from 192.168.20.4: bytes=32 time=1ms TTL=120
Ping statistics for 192.168.20.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
```

Option 2: DHCP

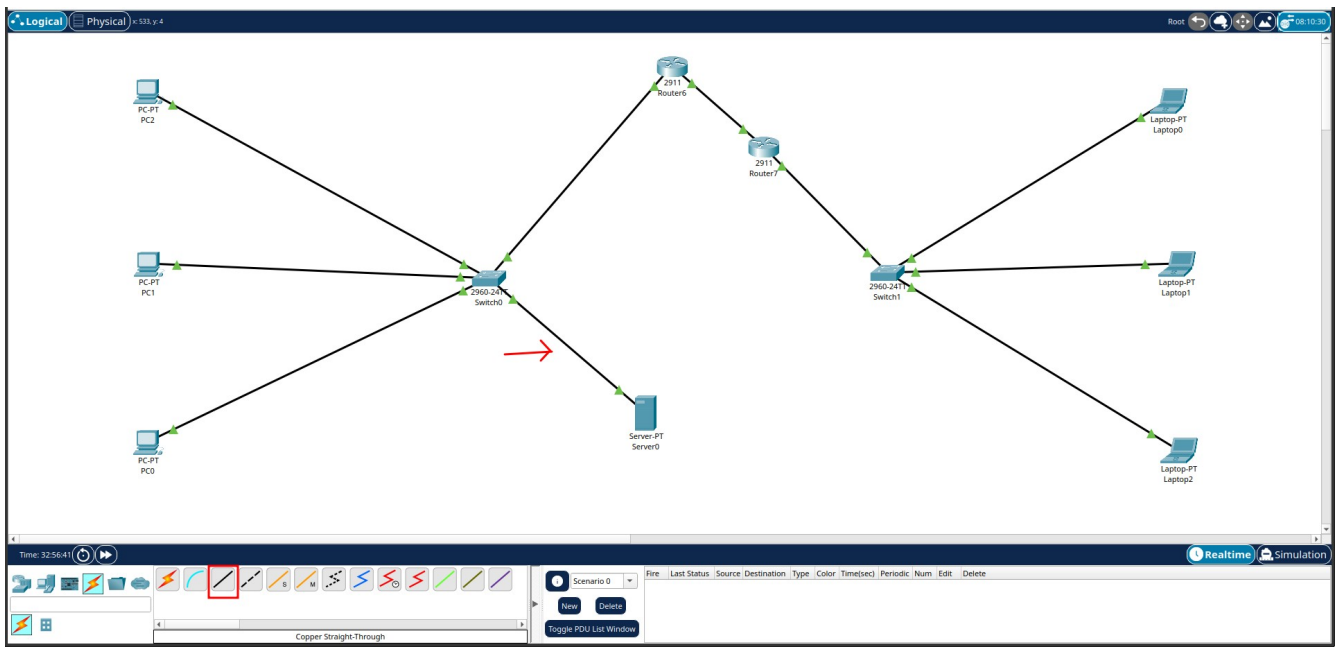
Step 1:

Install a Server



Step 2:

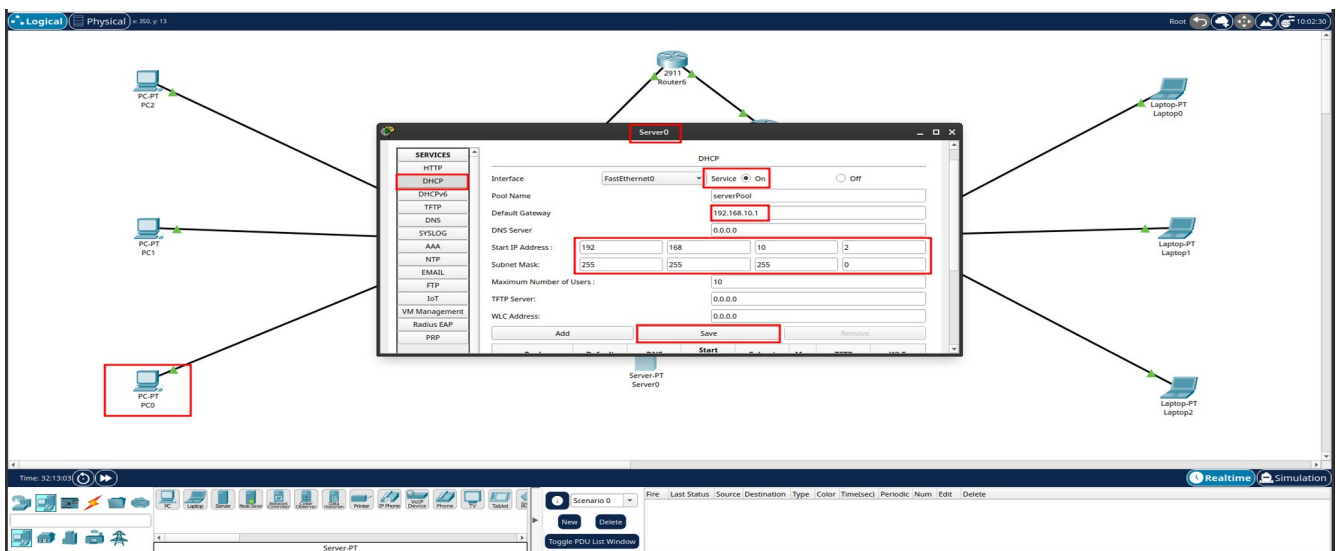
Connect the Server to switch0



Step 3:

Configure the server.

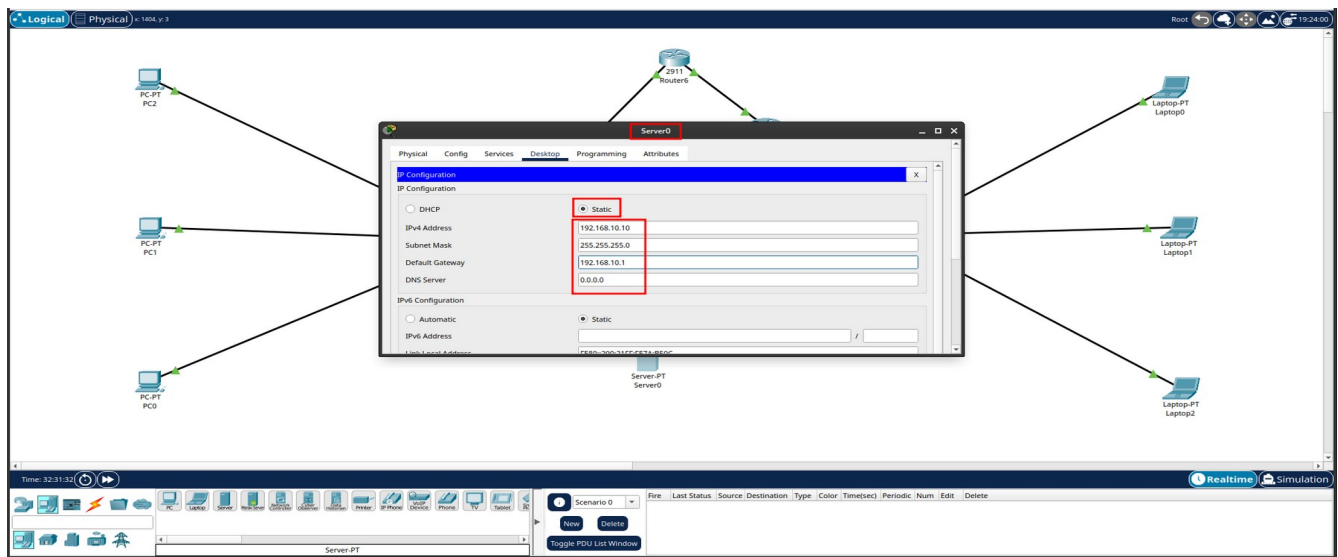
- Click on the “server”, Click on “service”, click on DHCP and turn service on.
- Enter the Gateway IP of switch0 which is connected to the server (192.168.10.1).
- Enter the start IP. In this case, I use PC0 (192.168.10.2)
- Enter subnet mask (255.255.255.0) and save



Step 4:

Assign IP to the server

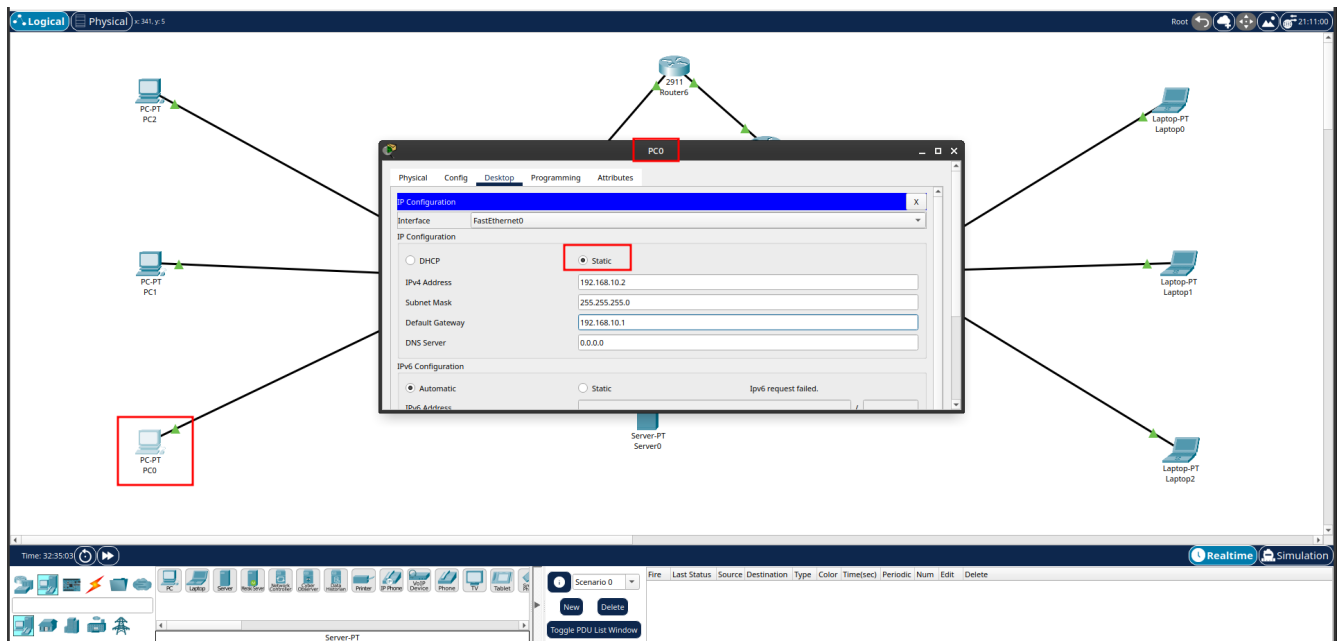
On the server, click on “Desktop”, assign IP (192.168.10.10) to the server and enter subnet and Gateway.



Step 5:

Testing Connectivity with DHCP

i. Testing with PC0, this is the position on the previous static configuration



ii. Now choosing the DHCP option by turning it on.

Result: Connectivity is successful.

Logical Physical 11:29:00

PC0

PC-PT PC2

PC-PT PC1

PC-PT PC0

Laptop-PT Laptop0

Laptop-PT Laptop1

Laptop-PT Laptop2

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.10.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static IPv6 request failed.

IPv6 Address

Link Local Address FE80::260:3EFF:FEDE:394C

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

☐ Top

Time: 33:03:12

Realtime Simulation

Scenario 0

Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

None Delete

Toggle PDU List Window

Copper Straight-Through